Model Translation



Model Translation

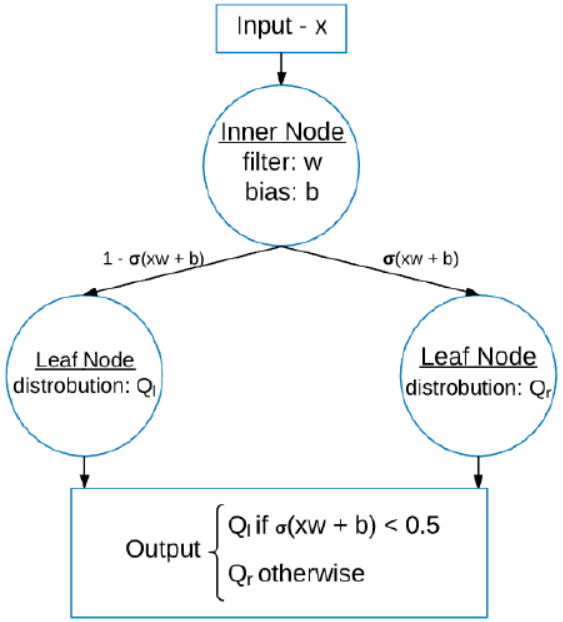
Translate (compress) the models into more interpretable ones

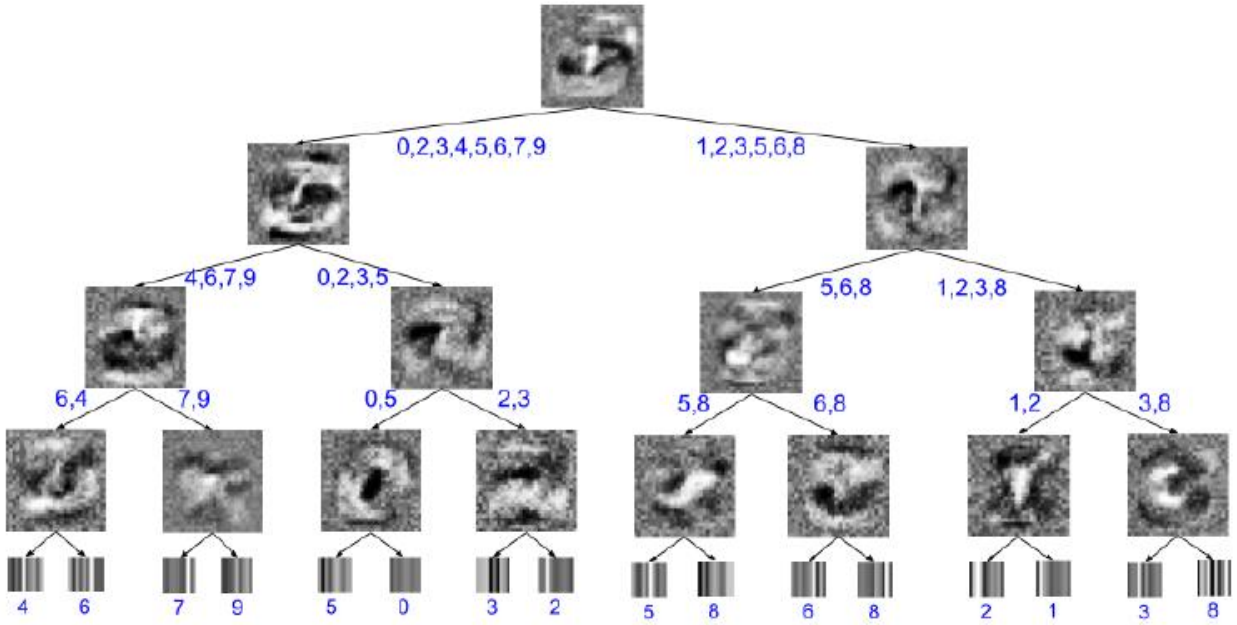
- Neural networks are hardly interpretable but decision trees are interpretable
- Neural networks learn more generalizable functions given certain amount of data than decision trees
- Can we translate neural networks into decision trees representing similar functions using unlabelled or even synthetic data?

- Chandra et al. “The Combination and Comparison of Neural Networks with Decision Trees for Wine Classification”, 2007
- Frost&Hinton “Distilling a Neural Network Into a Soft Decision Tree“, 2017
- Tan et al. “Transparent Model Distillation”, 2018



Translate (compress) the models into more interpretable ones





[Frost&Hinton "Distilling a Neural Network Into a Soft Decision Tree", 2017]

Content

- Curious cases of deep neworks
- Understanding how deep networks learn
- Curious cases of deep networks (cont.)
- Model Interpretability
  - Model Visualization
  - Model Translation
- Decision (prediction/estimation) Explainability
  - More motivation
  - Challenges
  - Types

